Canonical two-chunk planner user guide

Version 0.2.0. `two_chunk_planner` is a completely standalone, read-only canonical two-chunk planning tool. After installation it runs from any directory and needs no ULVZ project, SPECFEM worktree, or patch manifest.

1. Scope and limits

It plans only `NCHUNKS=2`, canonical 90°×90° geometry: AB is the central first chunk and AC is the supported-left second chunk. It does not support other widths, attachment sides, independent orientations, arbitrary two-chunk or multi-chunk topology. `general_two_chunk_mode_classification=B`.

The tool searches center latitude/longitude and `GAMMA_ROTATION_AZIMUTH`, classifies sources/receivers, audits paths and emits a reviewable parameter fragment. It never runs or replaces topology, mesher, database, Stacey, solver, waveform, or production boundary-return acceptance.

2. Install and start

Install a supplied wheel from any directory:

```
python -m pip install two_chunk_planner-0.2.0-py3-none-any.whl
```

Install from a source distribution or source directory:

```
python -m pip install two_chunk_planner-0.2.0.tar.gz
python -m pip install .
python -m pip install -e '.[phase-aware]'
```

The optional `phase-aware` extra provides ObsPy. Confirm geographic TauP support instead of assuming it:

```
python -c 'from obspy.geodetics.base import HAS_GEOGRAPHICLIB; print(HAS_GEOGRAPHICLIB)'
two-chunk-planner --help
python -m two_chunk_planner --help
```

For package development only, run without installing:

```
PYTHONPATH=src python -m two_chunk_planner --help
```

3. Quick standalone cases

The following fixtures are synthetic and not Kim/Song inputs. Set `P` to a copy of the package examples; every output directory must not exist.

```
P=/path/to/two_chunk_planner/examples
two-chunk-planner plan --cmtsolution "$P/geometry_only/DATA/CMTSOLUTION" \
  --stations "$P/geometry_only/DATA/STATIONS" --analysis-window 0 1900 \
  --latitude-range 0,0 --longitude-range 0,0 --gamma-range 0,0 --output geometry_plan
```

```
two-chunk-planner plan --cmtsolution "$P/phase_aware/DATA/CMTSOLUTION" \
  --stations "$P/phase_aware/DATA/STATIONS" --path-mode phase-aware \
  --phases Pdiff,Sdiff --taup-model prem --taup-resample --analysis-window 0 1900 \
  --latitude-range 0,0 --longitude-range 0,0 --gamma-range 0,0 --output phase_plan
```

```
two-chunk-planner plan --cmtsolution "$P/geometry_only/DATA/CMTSOLUTION" \
  --stations "$P/geometry_only/DATA/STATIONS" --par-file "$P/external_par_file/Par_file" \
  --analysis-window 0 1900 --latitude-range 0,0 --longitude-range 0,0 --gamma-range 0,0 \
  --output external_par_plan
```

4. Inputs and profile

Provide exactly one of `--cmtsolution` or `--source LAT LON DEPTH_KM`, and one of `--stations` or `--stations-csv`. Source, station and target YAML files may be at any readable path. Circle/polygon/corridor targets are optional.

`--par-file` is the only optional external configuration context. It may be at any path; it reads NEX/NPROC and relevant compatibility flags but is neither located through SPECFEM nor edited. If absent, planning defaults come from `src/two_chunk_planner/resources/canonical_profile_v1.json`: NEX=96 and the canonical geometry/provenance reference. These are planning defaults, not the user's production configuration. The profile never accesses a user SPECFEM tree, patch manifest, or source hash.

The English CLI reference and Chinese CLI reference are the complete, authoritative 35-option list. Use them for TauP, target, search, scoring, NEX/MPI and output details; this guide explains only common workflow choices.

5. Path modes and planning

`geometry-only` is the default and uses sampled surface great circles. `phase-aware` requests each requested phase×station with TauP; Pdiff/Sdiff geographic paths are validated. Strict coverage is default; partial coverage requires `--allow-partial-phase-coverage` and never substitutes a phase. TauP length is `taup_raypath_polyline_estimate`; CMB-near information is a sampled proxy. TauP is forbidden for boundary-return timing.

Search is deterministic coarse/local/final center-gamma refinement. Candidate geometry remains canonical; target coverage, external margins and endpoint margins are hard/transparent checks. NEX=96 at total ranks 2/8/12 is labelled `project_validated`; other compatible choices are not project-validated.

6. Outputs and verification boundary

Each successful run writes `candidates.json`, `candidates.csv`, `geometry_audit.json`, `boundary_time_audit.json`, `report.md`, `map.png`, `run_manifest.json`, and `recommended_Par_file.inc`.

All audits record `planner_mode=standalone`, `compatibility_profile_version`, `specfem_source_verified=false`, `accepted_patch_verified=false`, `production_configuration_verified=false`, `par_file_source`, `configuration_status`, and `verification_warnings`. These false values are not planning failures: independent planning deliberately does not inspect a user's SPECFEM source, installed patch, mesh, or runtime validation.

`recommended_Par_file.inc` is a suggested parameter fragment, not a complete or runnable Par_file. Review candidates, geometry audit, boundary-time audit, report and fragment before copying anything manually.

7. Complete workflow

1. Install the package and prepare source/stations.
2. Choose geometry-only screening or phase-aware review.
3. Optionally prepare a strict target YAML and/or external Par_file.
4. Run `plan` to a new output directory.
5. Review candidate ranking, classifications, margins, warnings and resource labels.
6. In a separate SPECFEM workflow, apply/verify the accepted patch, then validate mesh/topology, Stacey, solver, waveform and external returns.

8. Boundary time, license and scientific status

Boundary seconds are always advisory `heuristic_not_conservative` surface-arc proxies; `boundary_time_production_safe=false`. Sampling stability is `indeterminate` (Pdiff 0.130094%, Sdiff 0.076131% resample-length differences). Kim/Song exact reconstruction remains unavailable without author inputs.